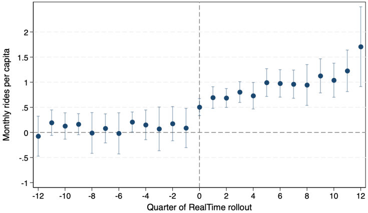
Comments on: “Navigating Change: Google Maps, Real-Time Information, and Transit Ridership” by R. Jerch, A. La Nauze, X. Peng

Matthew A. Turner

Brown University

July, 2026



- Unit is a transit district month.
- x axis is quarters from rollout of real time google maps transit features (2011-19).
- y axis riders per month per capita from NTD (BJS estimator $\Longrightarrow$ no omitted month).
- Treatment effect is about 1 ride month/pp.
- Base is about 7 rides pp/month, so this is a 14% increase.
- Federal budget for transit is about 50b a year.

Could this possibly be true?

At best, google maps cuts wait time to zero. Can we guess how important this would be?

- Value of wait time per minute 0.87(bus off-peak USD2002) from Parry and Small AER 2009.
- Elasticity of passenger demand wrt fare 0.80 (bus off-peak)
- Suppose (1) google maps decreases generalized cost of trip by 20% (2) fare and generalized cost elasticity are the same.
- $0.2 \times 0.8 = 0.16 \approx 0.14$. In the ballpark.
- Do this calculation more carefully.

Subways vs. Buses

- New York is about 70% of all subway rides. Top six subway cities are above 90%. Top six transit cities account for about 40% of all bus rides.
- Weighting by riders vs transit districts changes this from a paper about lots of little bus districts to a case study of NY (and maybe 5 other places).
- I think you are better set up to describe lots of little bus districts. Focus on this, drop big subway cities from main analysis and treat them in a little section at the end?
- Related #1. Except for NYC, I think buses carry at least as many riders as subways almost everywhere. Is figure A.III really just reporting on NY?
- Related #2. Are results for complicated/real time subsets both driven by NYC and other large districts? (more below).

Econometrics I

Recall the logic of the DiD/BJS estimator

- Potential outcomes for treated and untreated states,

$$Y_{it}(0), Y_{it}(1), t = 1, 2.$$

- Want

$$ATT = E(Y_{i2}(1) - Y_{i2}(0)|D_i = 1).$$

  ... but $Y_{i2}(0)|D_i = 1$ is never observed.
- BJS impute $Y_{i2}(0)$ from untreated units, usually

$$\widehat{Y_{i2}}(0) = \widehat{\alpha}_i + \widehat{\beta}_2 + \widehat{\gamma} x_{i2}.$$

- I can impute $\widehat{Y_{i2}}(0)$ from the price of fish and, as long as I maintain conditional parallel trends, the logic of the estimator goes through. However, (1) I am shifting from a non-parametric to a parametric imputation function, (2) the model of treated unit counterfactual untreated state is less credible a priori.

Econometrics II

Here is the paper's imputation function,

$$\widehat{Y_{it}}(0) = \eta_i + [\gamma_{dy} + \tau_{ds} + \rho_{iy}]$$
$$\eta_i = \text{City}, \ \gamma_{dy} = \text{Region Year}, \tau_{ds} = \text{Region Season},$$
$$\rho_{iy} = \text{City Year controls}, \ t, l, s = \text{Month, Quarter, Year}$$

Here is the conventional version

$$\widehat{Y_{it}}(0) = \eta_i + \beta_t$$

- The LHS has variation at the month level. Right is *years*. Why move away from non-parametric imputation?
- With BJS, calculate ATT one month and city at a time, and aggregate $\Longrightarrow$ the whole paper needs only one regression. Get subsample results by averaging a subset of unit ATT's.

What is real time information worth?

Try to tell us how important this is. How big a change would we need in fares or service quality to induce the same change in behavior? How much would this cost? How much does google maps cost? Does it make sense for transit districts to invest in providing even better information?

Summary

- Estimate a huge effect of an information treatment.
- Looks like it could be consistent with other things we know, but make sure.
- Econometrics are probably OK, but make sure.
- I think this is a paper about lots of little bus districts. This is OK, buses are the most important transit mode almost everywhere. But try to be clearer about this.